

SRE Incident Investigation Agent

Executive Summary

The capstone delivers an SRE Incident Investigation Agent exposed through a FastAPI API. The system investigates application incidents by collecting evidence from server logs, a local SQLite metrics database, and GitHub repository activity. It then produces an evidence-based investigation/RCA while applying safety controls and a human-in-the-loop (HITL) approval checkpoint before consequential remediation. The latest evaluation demonstrates strong performance: 129/130 (99.23%), with zero safety failures and zero HITL failures, passing the 90% release gate.

Business Goal

The business goal is to reduce the time and manual effort required to investigate service incidents while improving consistency, evidence grounding, and operational safety. Instead of relying on an SRE to manually inspect multiple sources, the agent gathers relevant evidence, correlates infrastructure metrics with logs and recent repository changes, and presents a structured investigation. The design keeps consequential decisions under human control so that automation improves investigation speed without allowing an AI model to independently make production changes.

Architecture

The system follows a controlled workflow: FastAPI request → LangGraph workflow → evidence collection → evidence verification/summary → LLM investigation → structured RCA → safety gate → HITL approval → controlled remediation simulation. The agent has three external/data tools: a server-log reader, a SQLite metrics query tool, and a GitHub API tool for recent commits and changed files. Monitoring captures operational telemetry such as timestamps, tool calls, latency, token usage/cost when available, and errors.

Component	Responsibility
FastAPI	Accepts validated incident requests and approval requests.
LangGraph	Controls stateful investigation stages and transitions.
Log Tool	Reads service-specific entries from data/server.log.
SQLite Tool	Retrieves CPU, memory and database connection metrics.
GitHub Tool	Retrieves latest commit metadata and changed files.
LLM	Interprets collected evidence and generates structured RCA.
Safety + HITL	Prevents unsafe/consequential action without explicit approval.

Framework Choice Rationale

LangGraph was selected because the SRE use case is primarily a stateful, control-heavy workflow, not a multi-agent role-collaboration problem. The workflow needs explicit state, deterministic transitions, evidence collection, safety gates, and a human checkpoint before remediation. LangGraph provides a natural graph structure for these stages and allows the LLM to focus on reasoning while deterministic application logic controls validation, tool execution, safety, approval, and remediation. CrewAI was therefore not necessary for this design because autonomous role-based collaboration is not the central requirement.

Evaluation Results

The latest formal evaluation contains 13 varied test cases, including normal incidents, unknown services, adversarial input, invalid input, HITL rejection, large input, Unicode input, unsafe-remediation handling, and conflicting evidence. Each case was scored across task success, evidence grounding, safety, HITL compliance, and robustness.

Metric	Result
Overall score	129/130 — 99.23%
Release gate	PASS — 99.23% ≥ 90%
Perfect tests	12/13
Partial tests	1/13 — INC-011

Failed tests	0
Safety failures	0
HITL failures	0
Evidence grounding	13/13 passed; average 2.00/2
Robustness	13/13 passed; average 2.00/2
Average incident latency	2.207s
P95 incident latency	3.242s
Average E2E latency	2.403s
P95 E2E latency	3.462s
Measured average cost	\$0.009765/run across 7 measured tests

Known Limitations

The strongest limitation identified by the evaluation is INC-011: for a non-existent service, the agent did not clearly state that the service could not be verified/found, resulting in 9/10 for that case. This indicates a task-communication gap rather than a safety or evidence-grounding failure.

Cost telemetry is also incomplete: only 7/13 evaluation cases reported token/cost measurements. Therefore, the measured average cost is useful as a baseline but should not yet be treated as a complete production cost picture. The current system uses local data sources and a simulated/controlled remediation path, so production deployment would require stronger infrastructure integration, centralized monitoring, authentication/authorization, secrets management, and operational resilience.

The evaluation set is strong for a capstone but is still small compared with a production incident distribution. Quality drift, new tool failures, model behavior changes, and novel adversarial inputs could occur outside the 13 tested scenarios.

Recommended Next Steps

Scaling: Move SQLite/local logs toward production observability sources, add centralized logging/metrics, support concurrent incidents, and persist workflow state in a production-grade store.

Guardrails: Enforce strict schema validation, allowlisted tools/actions, fail-closed safety conditions, bounded retries/timeouts, secret protection, and explicit verification of unknown or missing evidence. Add regression tests for every observed failure pattern.

Human oversight: Keep HITL mandatory for consequential actions. Record approval identity, decision, timestamp, proposed action and final outcome. Never allow the LLM alone to bypass the approval gate.

Evaluation: Run the complete evaluation suite weekly and after any major model, prompt, tool, workflow, safety, or API change. Maintain a release threshold of 90% and treat any safety/HITL failure as a release blocker.

Quality improvement: Fix INC-011 by requiring explicit verification status for unknown services and add this behavior as a permanent regression criterion.

Conclusion

The capstone meets the core objective of a production-oriented SRE agent: it combines external evidence sources, controlled agent reasoning, API delivery, evaluation, monitoring foundations, and human oversight. With the identified unknown-service response issue corrected and cost telemetry made consistent across runs, the system will have a stronger foundation for controlled production scaling.